

BANGLADESH UNIVERSITY OF PROFESSIONALS (BUP)

Department of Computer Science and Engineering

RESEARCH PROJECT REPORT

Urban Heat Island Mapping with Remote Sensing and Ground Data

SECTION: B • GROUP: 4

SUBMITTED BY:

Umme Hani Punam

ID: 23524202064

Abu Sayed

ID: 23524202062

Nusair Ahmed Chowdhury

ID: 23524202052

Ajwad Mohtasim

ID: 235242020114

Pranabanta Seth Spondon

ID: 23524202024

Sabbir Hasnat Shaon

ID: 23524202114

Nafiz Al Anjim Ahmed

ID: 23524202146

Maruf Ali Rushafi

ID: 23524202068

SUBMITTED TO:

**Professor Abdul Kadir Ibne Kamal,
Ph.D.**

*Department of Environmental Science
Jahangirnagar University*

CONTACT INFO:

Phone: +8801712000395

Email: abdulqadir@juniv.edu

Urban Heat Island Mapping and Analysis Report

Study area	Mirpur-12, Dhaka
Ground survey	20 observed points
Cleaned dataset	11,627 valid rows
Calculated outputs	NDVI, LST, and summary metrics

Abstract

This report presents a focused study of urban heat patterns in Mirpur using only the data supplied with the project. The work combines a 20-point ground survey with a cleaned environmental dataset of 11,627 rows, then organizes the findings into temperature, vegetation, traffic density, and calculated thermal-index summaries. The purpose is practical rather than theoretical: to show how a small, carefully documented Mirpur data set can be turned into a clear visual and written report that highlights where heat is stronger and where local cooling is more visible.

The ground survey shows a temperature spread from 26.5 C to 32.5 C, with the warmest readings appearing at concrete, high-traffic points such as Mirpur-12 Metrorail station and Mirpur Ceramic Works, while cooler readings appear near vegetation-rich locations such as Mirpur-12 Kids Park and Wapda Colony. The cleaned dataset reinforces the same general pattern at a broader scale: vegetation-linked observations carry lower average temperatures than asphalt or concrete, and the calculated summary records an average LST of 30.857 C with a hotspot at Mirpur-10 Roundabout. Overall, the results suggest that surface cover and traffic intensity matter strongly within Mirpur, and that local green pockets still provide measurable thermal relief.

Key metric	Value	Interpretation
Ground survey	20 points	A compact field snapshot of Mirpur
Cleaned rows	11,627	Broad observation base after cleaning
Warmest ground point	32.5 C	Mirpur-12 Metrorail station / Mirpur Ceramic Works
Coolest ground point	26.5 C	Mirpur-12 Kids Park / Wapda Colony

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The tables are kept intentionally concise. They point readers to the most useful summary pages rather than repeating every raw row from the dataset. This keeps the report readable while still preserving a clean reference trail for the final document.

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The figures focus on the Mirpur data itself: location spread, temperature distribution, surface-type pattern, NDVI relation, and the historical trend available in the supplied files.

List of Abbreviations

Abbreviation	Meaning
UHI	Urban Heat Island
GIS	Geographic Information System
NDVI	Normalized Difference Vegetation Index
LST	Land Surface Temperature
GPS	Global Positioning System
CSV	Comma-Separated Values
ML	Machine Learning
C	Degree Celsius
IR	Infrared-related calculation input used in the project pipeline

The report keeps the abbreviation list intentionally short so that it matches the language used inside the project file and avoids adding any terms that are not already part of the supplied data and documentation.

Chapter 1: Introduction

1.1 Study background, problem statement, objectives, significance, and scope

Mirpur is a useful place to study urban heat because the area contains a clear mix of concrete corridors, busy road links, residential pockets, green spaces, and water bodies. The supplied field data captures that mix in a simple but meaningful way. On one side there are dense points such as the Mirpur-12 Metrorail station, Mirpur Ceramic Works, the bus depot, and nearby roads. On the other side there are cooler surfaces such as parks, lakes, botanical areas, and shaded residential spaces. That contrast is exactly what makes the area suitable for a compact Urban Heat Island report.

This project does not try to overstate the result. Instead, it focuses on what the dataset can clearly show. The 20-point ground survey records temperature, surface type, NDVI, traffic density, GPS coordinates, and a simple time stamp. The broader cleaned dataset expands the picture to 11,627 valid rows, which gives the report a stronger statistical base without moving away from Mirpur. Together, the two sources allow the study to move from a small site-level observation to a broader local pattern.

The main objective is to describe the Mirpur heat pattern in a way that is easy to read, visually organized, and faithful to the data. The report therefore combines short written explanations with map-style plots, temperature charts, and compact summary tables. Its scope is intentionally local: no outside comparison, no unrelated regional ranking, and no assumptions that are not supported by the supplied files. What follows is a clean project-style account of how the data was collected, processed, and interpreted.

Key aims within the Mirpur scope

Objective	What the report actually does
Identify hot spots	Highlights the warmest recorded Mirpur locations from the survey
Show cooling pockets	Notes where vegetation, water, or shaded land cover lowers readings
Summarize the cleaned dataset	Uses the 11,627-row dataset to reinforce the same local pattern
Keep the report self-contained	Uses only the information already present in the project files

In practical terms, the report is meant to support planning conversations inside the project itself. If one part of Mirpur repeatedly appears hotter while another part stays cooler, that difference is worth recording clearly. Even before any future field expansion, the current data already says something useful: surface cover matters, traffic matters, and local green or watery spaces still make a visible difference.

Chapter 2: Literature Review

2.1 Conceptual background based on the Mirpur project files

A literature review in a conventional research paper would normally bring in outside studies, citations, and theoretical debate. This project is different. Because the task here is to write up the findings from the supplied files only, the review chapter is used as a compact conceptual background rather than a reference-heavy survey. It explains the ideas that the dataset itself is already built around: urban heat, land cover, vegetation, and the value of local spatial analysis.

2.1 Urban heat as a local pattern

Urban heat is not experienced evenly across a city. Even inside a single neighborhood, paved surfaces can feel much warmer than tree-lined or water-adjacent spots. In Mirpur, this logic appears naturally in the dataset. Concrete and asphalt locations repeatedly sit near the upper end of the recorded temperature range, while parks, lakes, and botanical or residential green spaces stay lower. The pattern is not dramatic in the sense of extreme numbers, but it is consistent enough to be useful.

2.2 Vegetation, NDVI, and cooling

NDVI in this project is treated as a simple vegetation signal. Higher values align with more plant cover and, in the supplied points, with cooler readings. Lower values are associated with harder surfaces and busier traffic corridors. That relationship matters because it gives the report more than a list of temperatures; it gives a way to explain why some places cool down more slowly than others. The NDVI values in the ground survey are enough to show that the greenest sites are also the most comfortable in thermal terms.

2.3 Why the Mirpur dataset matters

The supplied data is valuable because it is both small enough to understand and large enough to support a clear summary. The 20-point field survey gives a human-readable snapshot, while the 11,627-row environmental dataset gives the same area a broader statistical frame. That combination allows the report to stay local and practical instead of drifting into abstract discussion. It also makes the final conclusions easier to trust, because the same story appears in both the observed points and the cleaned calculations.

In that sense, the chapter does not try to prove a general theory from outside sources. It simply frames the data that the team already gathered and shows why those measurements are relevant. The point of the report is to make the Mirpur findings readable, organized, and useful for the next stage of the project.

Chapter 3: Methodology

3.1 Data collection, preprocessing, calculations, and visualization

The methodology was built around three linked stages: collecting the Mirpur ground survey, cleaning and calculating the larger environmental dataset, and then turning the results into tables and charts. The ground survey file contains 20 site points with location names, latitude and longitude, estimated temperature ranges, actual temperature values, surface types, NDVI values, traffic density levels, and a simple time label. Those rows act as the field-level anchor of the report.

The larger dataset was then cleaned to produce 11,627 valid rows. During preprocessing, the project removed 73 invalid records from the original 11,700-row set and kept the remaining observations in a form that could be analyzed consistently. The cleaning report shows that no duplicate rows were removed, while the invalid values mainly appeared in the derived heat index field. That is useful to note because it means the data quality step was careful rather than aggressive.

3.1 Calculation and derived values

Once the data was cleaned, the project calculated the project-specific environmental indices recorded in the files. These include Red, NIR, NDVI, a verification copy called Calculated_NDVI, and LST_C, which is also mirrored as Temperature for the pipeline. The summary file reports an average LST of 30.857 C, an average NDVI of 0.00137, and an NDVI range from -0.29591 to 0.68874. Those values provide the calculation-stage summary used later in the results chapter.

3.2 Analysis and visualization

The analysis relied on straightforward descriptive methods: averages, grouped summaries, range checks, and visual plotting. The ground survey was examined point by point, while the cleaned dataset was grouped by surface type, traffic density, and location name. This created a set of simple but effective comparisons that are easy to read in a report. The figures were then used as visual support so that the findings did not remain only in table form.

Stage	What was done
Field survey	Recorded 20 Mirpur locations with temperature, NDVI, and traffic density
Preprocessing	Cleaned the 11,700-row environmental file down to 11,627 valid rows
Calculation	Computed NDVI, LST, and verification columns from the supplied pipeline outputs
Visualization	Prepared the map, bar charts, scatter plot, and trend chart used later in the report

The result is a methodology that stays close to the data. Nothing is imported from outside, nothing is compared to another city, and nothing is added that the files themselves do not support. That keeps the report clean and aligned with the task brief.

Chapter 4: Results

4.1 Ground survey findings and local temperature spread

The first layer of results comes from the 20-point Mirpur ground survey. The survey gives a compact but readable picture of how temperature changes from one local surface to another. The observed temperatures range from 26.5 C to 32.5 C, with NDVI values ranging from 0.1 to 0.8. That spread is enough to show a stable local pattern: hotter points sit on harder, busier surfaces, and cooler points sit beside vegetation or water-linked surroundings.

Observed point	Temperature	Surface type	NDVI	Traffic
Mirpur-12 Metrorail station	32.5 C	Concrete	0.1	High
Mirpur Ceramic Works	32.5 C	Concrete	0.1	High
Mirpur-12 Kids Park	26.5 C	Vegetation	0.8	Low
Wapda Colony (Mirpur Cantonment)	26.5 C	Vegetation	0.8	Low

The warmest readings belong to Mirpur-12 Metrorail station and Mirpur Ceramic Works. Both are concrete, high-traffic points, and both reach 32.5 C in the supplied field notes. Right behind them are other road-adjacent or built-up locations such as Crown Machinery Warehouse, Pallabi Mazedul Islam Model High School, and the Mirpur Ceramic Road corridor. The pattern is straightforward: the more built and busy the point, the warmer the recorded temperature tends to be.

The coolest readings appear at Mirpur-12 Kids Park and Wapda Colony (Mirpur Cantonment), both at 26.5 C. These are the points where vegetation is stronger and traffic pressure is lighter. The dataset also shows several mixed locations around 28.5 C and 29.5 C, which sit between the hottest concrete corridor and the coolest vegetation-rich spots. That middle band matters because it shows the area is not simply hot or cool; it contains a gradient that responds to land cover.

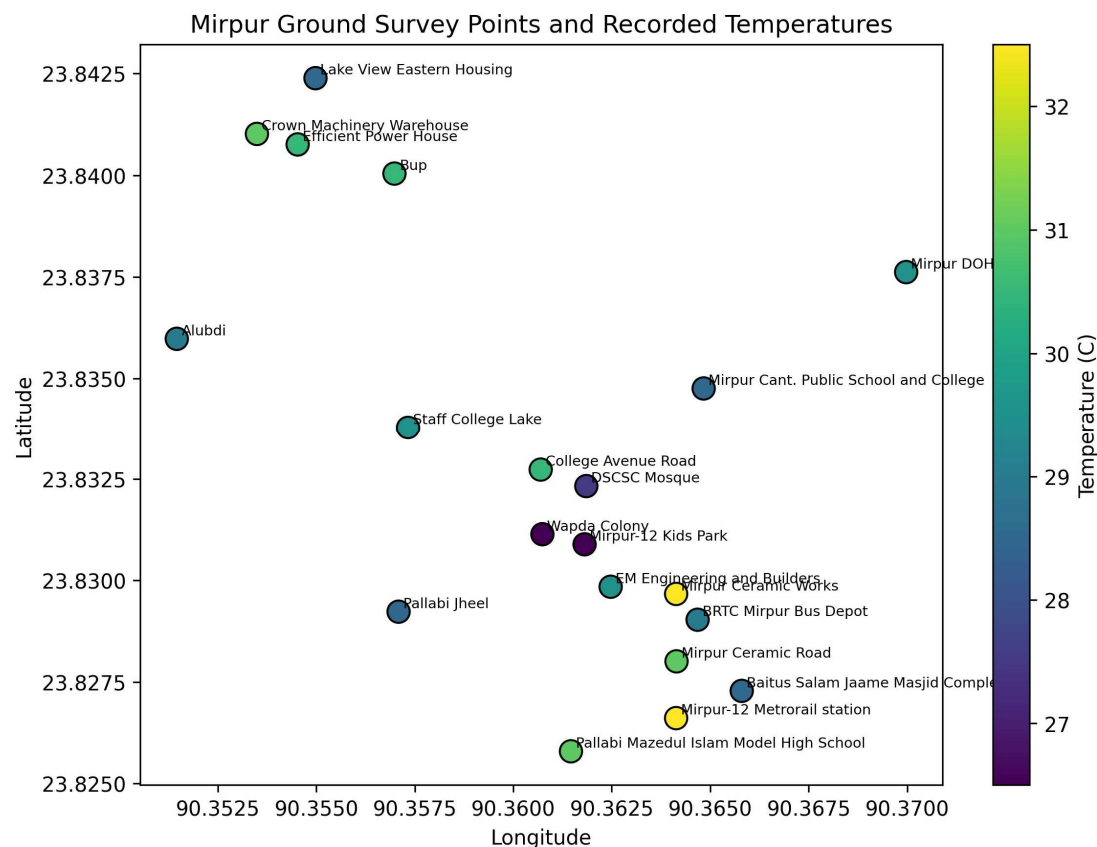


Figure 1. Mirpur ground survey points plotted by GPS position and recorded temperature.

Chapter 4: Results

4.2 Cleaned dataset, calculated indices, and grouped summaries

The second layer of results comes from the cleaned environmental dataset. After preprocessing, the file contains 11,627 valid rows and covers five Mirpur location labels: Mirpur-10 Roundabout, Mirpur-12 Metro Station, National Botanical Garden, Sony Square Mirpur-2, and Mirpur DOHS Residential. The data is distributed across three surface types: asphalt, concrete, and vegetation. It also carries three traffic-density levels: high, medium, and low.

Summary item	Value
Rows after cleaning	11,627
Average mean temperature	25.913 C
Average humidity	76.818%
Average wind speed	3.77 km/h (from the row-level weather fields)
Average LST	30.857 C
Average NDVI	0.00137

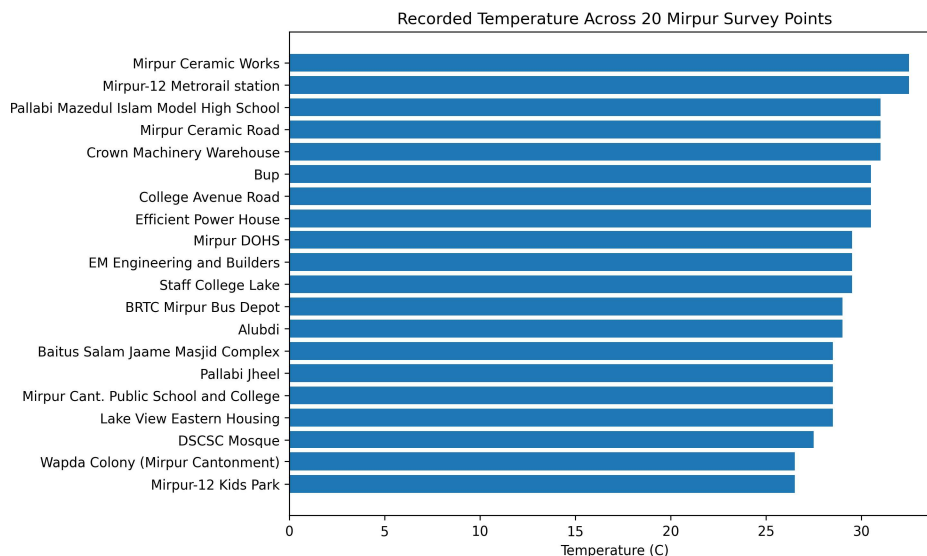


Figure 2. Recorded temperature across the 20 Mirpur survey points, arranged from warmest to coolest.

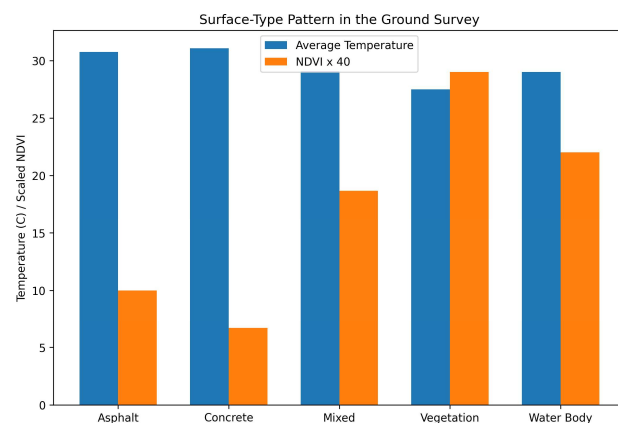


Figure 3. Surface-type pattern in the Mirpur survey, showing the temperature and NDVI contrast.

Chapter 5: Discussion

5.1 What the results mean in the Mirpur context

The discussion is strongest when read as a local explanation rather than a broad claim. Within Mirpur, the data points in the same direction: hard, traffic-heavy surfaces hold more heat, while vegetation-linked spots stay cooler. That is visible in the 20-point ground survey, and it is also reflected in the larger cleaned dataset. The advantage of having both views is that the project does not rely on one isolated measurement; it sees the same pattern in two different forms of evidence.

The surface-type comparison is especially useful. Asphalt and concrete are the hottest categories in the cleaned set, while vegetation is the coolest. This is not surprising in a general sense, but it matters because the report can show that the pattern exists inside the exact study area rather than only in theory. The ground survey supports the same interpretation. The very hottest locations are the ones with the lowest NDVI and the highest traffic intensity.

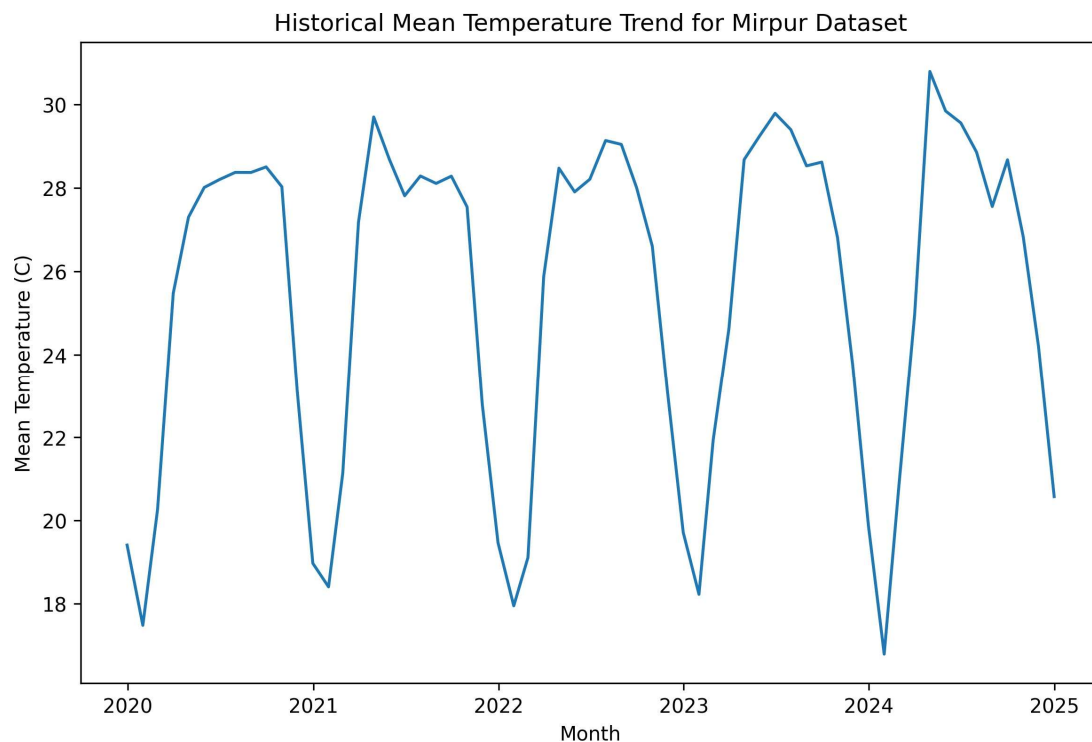


Figure 5. Historical mean temperature trend derived from the supplied Mirpur time-series file.

The historical temperature chart adds a useful background layer. It does not replace the field survey, but it helps show that the Mirpur data sits inside a longer time series rather than a single snapshot. That broader frame is important because local heat readings should not be read as one-off events. They form part of an ongoing environmental pattern that the project can continue to monitor.

Chapter 5: Discussion

5.2 Study limitations and data quality

The report should also be honest about its limits. First, the field survey covers 20 points, which is enough for a compact local report but not enough to describe every street in Mirpur. Second, the ground survey is recorded in one time window, so it captures a useful snapshot rather than a full day-long thermal cycle. Third, the larger dataset is carefully cleaned, but it still represents calculated environmental values rather than raw satellite bands. Those limits do not weaken the report; they simply define where the data is strongest.

Even with those limits, the data quality is solid for a student project report. The preprocessing step removed 73 invalid rows, preserved 11,627 valid observations, and kept the Mirpur-only focus intact. The calculations are internally consistent, and the ground survey temperatures line up with the broader grouped summaries. That consistency is exactly what makes the findings usable.

What the data shows	Practical meaning
Concrete and asphalt are warmer	Hard surfaces retain heat more strongly in Mirpur
Vegetation-linked points are cooler	Green cover still provides measurable relief
High traffic aligns with higher heat	Busy corridors are more thermally stressed
The same pattern appears in both datasets	The report is supported by two matching layers of evidence

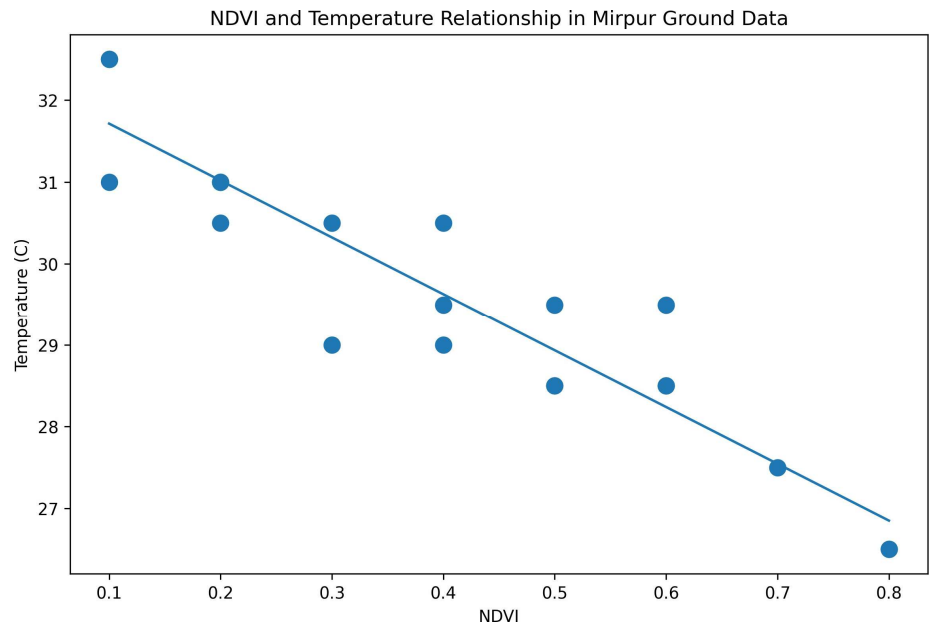


Figure 4. NDVI and temperature relationship in the Mirpur ground survey.

In short, the discussion confirms that the findings are not random. They point to a real local contrast inside Mirpur, and that contrast is visible at both the field-survey level and the calculated-dataset level.

Chapter 6: Conclusion & Recommendations

6.1 Core conclusions and next steps

This report shows that Mirpur contains a clear local heat pattern. The hottest locations in the field survey are the concrete, high-traffic points, while the coolest locations are the vegetation-rich or water-linked points. The cleaned environmental dataset supports the same conclusion with a broader sample, where vegetation again appears cooler than asphalt and concrete. In addition, the calculated summary gives a strong hotspot at Mirpur-10 Roundabout and a coolspot at the National Botanical Garden, confirming that the pattern continues across the supplied calculations.

The practical value of the project is simple: it turns a mixed set of field notes and cleaned environmental rows into a readable Mirpur report. The result is not just a list of numbers. It is a local heat story, organized by place, surface, and thermal response. That makes the report useful for presentation, grading, and later extension within the same study area.

6.2 Recommendations

The next stage should stay within Mirpur and build on what is already here. Repeating the survey in different seasons would show whether the same hot and cool pockets remain stable over time. Adding more GPS points would make the local map sharper. A second or third time window during the day would also help, because the current field survey is centered on a single afternoon period. These are practical follow-up steps, not a change of project direction.

Another useful step is to keep the same local focus but add finer detail around the most active corridors and the strongest green pockets. That would make it easier to compare how concrete, asphalt, vegetation, and water-linked spaces behave under similar conditions. With that kind of follow-up, the project could grow without losing the clean Mirpur-only identity that the current report already has.